```python
import pandas as pd
import numpy as np
df =pd.read_csv('loan_data_set.csv')
df
```

{"columns":[{"name":"index","rawType":"int64","type":"integer"},
{"name":"Loan_ID","rawType":"object","type":"string"},
{"name":"Gender","rawType":"object","type":"unknown"},
{"name":"Married","rawType":"object","type":"unknown"},
{"name":"Dependents","rawType":"object","type":"unknown"},
{"name":"Education","rawType":"object","type":"string"},
{"name":"Self_Employed","rawType":"object","type":"unknown"},
{"name":"ApplicantIncome","rawType":"int64","type":"integer"},
{"name":"CoapplicantIncome","rawType":"float64","type":"float"},
{"name":"LoanAmount","rawType":"float64","type":"float"},
{"name":"Loan_Amount_Term","rawType":"float64","type":"float"},
{"name":"Credit_History","rawType":"float64","type":"float"},
{"name":"Property_Area","rawType":"object","type":"string"},
{"name":"Loan_Status","rawType":"object","type":"string"}],"ref":"6ed3a80d-3e3b-4d70-94ba-0a83e0b4bdc5","rows":
[["0","LP001002","Male","No","0","Graduate","No","5849","0.0",null,"360.0","1.0","Urban","Y"],
["1","LP001003","Male","Yes","1","Graduate","No","4583","1508.0","128.0","360.0","1.0","Rural","N"],
["2","LP001005","Male","Yes","0","Graduate","Yes","3000","0.0","66.0","360.0","1.0","Urban","Y"],["3","LP001006","Male","Yes","0","Not Graduate","No","2583","2358.0","120.0","360.0","1.0","Urban","Y"],
["4","LP001008","Male","No","0","Graduate","No","6000","0.0","141.0","360.0","1.0","Urban","Y"],
["5","LP001011","Male","Yes","2","Graduate","Yes","5417","4196.0","267.0","360.0","1.0","Urban","Y"],["6","LP001013","Male","Yes","0","Not Graduate","No","2333","1516.0","95.0","360.0","1.0","Urban","Y"],
["7","LP001014","Male","Yes","3+","Graduate","No","3036","2504.0","158.0","360.0","0.0","Semiurban","N"],
["8","LP001018","Male","Yes","2","Graduate","No","4006","1526.0","168.0","360.0","1.0","Urban","Y"],
["9","LP001020","Male","Yes","1","Graduate","No","12841","10968.0","349.0","360.0","1.0","Semiurban","N"],
["10","LP001024","Male","Yes","2","Graduate","No","3200","700.0","70.0","360.0","1.0","Urban","Y"],
["11","LP001027","Male","Yes","2","Graduate",null,"2500","1840.0","109.0","360.0","1.0","Urban","Y"],
["12","LP001028","Male","Yes","2","Graduate","No","3073","8106.0","200.0","360.0","1.0","Urban","Y"],
["13","LP001029","Male","No","0","Graduate","No","1853","2840.0","114.0","360.0","1.0","Rural","N"],
["14","LP001030","Male","Yes","2","Graduate","No","1299","1086.0","17.0","120.0","1.0","Urban","Y"],
["15","LP001032","Male","No","0","Graduate","No","4950","0.0","125.0","360.0","1.0","Urban","Y"],["16","LP001034","Male","No","1","Not

Graduate","No","3596","0.0","100.0","240.0",null,"Urban","Y"],
["17","LP001036","Female","No","0","Graduate","No","3510","0.0","76.0"
,"360.0","0.0","Urban","N"],["18","LP001038","Male","Yes","0","Not
Graduate","No","4887","0.0","133.0","360.0","1.0","Rural","N"],
["19","LP001041","Male","Yes","0","Graduate",null,"2600","3500.0","115
.0",null,"1.0","Urban","Y"],["20","LP001043","Male","Yes","0","Not
Graduate","No","7660","0.0","104.0","360.0","0.0","Urban","N"],
["21","LP001046","Male","Yes","1","Graduate","No","5955","5625.0","315
.0","360.0","1.0","Urban","Y"],["22","LP001047","Male","Yes","0","Not
Graduate","No","2600","1911.0","116.0","360.0","0.0","Semiurban","N"],
["23","LP001050",null,"Yes","2","Not
Graduate","No","3365","1917.0","112.0","360.0","0.0","Rural","N"],
["24","LP001052","Male","Yes","1","Graduate",null,"3717","2925.0","151
.0","360.0",null,"Semiurban","N"],
["25","LP001066","Male","Yes","0","Graduate","Yes","9560","0.0","191.0
","360.0","1.0","Semiurban","Y"],
["26","LP001068","Male","Yes","0","Graduate","No","2799","2253.0","122
.0","360.0","1.0","Semiurban","Y"],
["27","LP001073","Male","Yes","2","Not
Graduate","No","4226","1040.0","110.0","360.0","1.0","Urban","Y"],
["28","LP001086","Male","No","0","Not
Graduate","No","1442","0.0","35.0","360.0","1.0","Urban","N"],
["29","LP001087","Female","No","2","Graduate",null,"3750","2083.0","12
0.0","360.0","1.0","Semiurban","Y"],
["30","LP001091","Male","Yes","1","Graduate",null,"4166","3369.0","201
.0","360.0",null,"Urban","N"],
["31","LP001095","Male","No","0","Graduate","No","3167","0.0","74.0","
360.0","1.0","Urban","N"],
["32","LP001097","Male","No","1","Graduate","Yes","4692","0.0","106.0"
,"360.0","1.0","Rural","N"],
["33","LP001098","Male","Yes","0","Graduate","No","3500","1667.0","114
.0","360.0","1.0","Semiurban","Y"],
["34","LP001100","Male","No","3+","Graduate","No","12500","3000.0","32
0.0","360.0","1.0","Rural","N"],
["35","LP001106","Male","Yes","0","Graduate","No","2275","2067.0",null
,"360.0","1.0","Urban","Y"],
["36","LP001109","Male","Yes","0","Graduate","No","1828","1330.0","100
.0",null,"0.0","Urban","N"],
["37","LP001112","Female","Yes","0","Graduate","No","3667","1459.0","1
44.0","360.0","1.0","Semiurban","Y"],
["38","LP001114","Male","No","0","Graduate","No","4166","7210.0","184.
0","360.0","1.0","Urban","Y"],["39","LP001116","Male","No","0","Not
Graduate","No","3748","1668.0","110.0","360.0","1.0","Semiurban","Y"],
["40","LP001119","Male","No","0","Graduate","No","3600","0.0","80.0","
360.0","1.0","Urban","N"],
["41","LP001120","Male","No","0","Graduate","No","1800","1213.0","47.0
","360.0","1.0","Urban","Y"],
["42","LP001123","Male","Yes","0","Graduate","No","2400","0.0","75.0",
"360.0",null,"Urban","Y"],

```
["43","LP001131","Male","Yes","0","Graduate","No","3941","2336.0","134.0","360.0","1.0","Semiurban","Y"],
["44","LP001136","Male","Yes","0","Not Graduate","Yes","4695","0.0","96.0",null,"1.0","Urban","Y"],
["45","LP001137","Female","No","0","Graduate","No","3410","0.0","88.0",null,"1.0","Urban","Y"],
["46","LP001138","Male","Yes","1","Graduate","No","5649","0.0","44.0","360.0","1.0","Urban","Y"],
["47","LP001144","Male","Yes","0","Graduate","No","5821","0.0","144.0","360.0","1.0","Urban","Y"],
["48","LP001146","Female","Yes","0","Graduate","No","2645","3440.0","120.0","360.0","0.0","Urban","N"],
["49","LP001151","Female","No","0","Graduate","No","4000","2275.0","144.0","360.0","1.0","Semiurban","Y"]],"shape":{"columns":13,"rows":614}}
```

```
df.isnull().sum()
```

{"columns":[{"name":"index","rawType":"object","type":"string"},{"name":"0","rawType":"int64","type":"integer"}],"ref":"02b4fb1a-3487-4932-8e9b-dd8ea4858c8c","rows":[["Loan_ID","0"],["Gender","13"],["Married","3"],["Dependents","15"],["Education","0"],["Self_Employed","32"],["ApplicantIncome","0"],["CoapplicantIncome","0"],["LoanAmount","22"],["Loan_Amount_Term","14"],["Credit_History","50"],["Property_Area","0"],["Loan_Status","0"]],"shape":{"columns":1,"rows":13}}

```
# Method 1: Deleting rows with missing values
df_deleting = df.copy()
df_deleting.dropna(inplace=True)
df_deleting.isnull().sum()
```

{"columns":[{"name":"index","rawType":"object","type":"string"},{"name":"0","rawType":"int64","type":"integer"}],"ref":"51bcf869-6875-4878-8408-c9aaefe3deb2","rows":[["Loan_ID","0"],["Gender","0"],["Married","0"],["Dependents","0"],["Education","0"],["Self_Employed","0"],["ApplicantIncome","0"],["CoapplicantIncome","0"],["LoanAmount","0"],["Loan_Amount_Term","0"],["Credit_History","0"],["Property_Area","0"],["Loan_Status","0"]],"shape":{"columns":1,"rows":13}}

```
# Method 2: Imputation using mean/median/mode
df_mean = df.copy()
for col in df_mean.columns:
  if df_mean[col].isnull().sum() > 0:
    if df_mean[col].dtype in ['int64', 'float64']:
      df_mean[col].fillna(df_mean[col].mean(), inplace=True)
    else:
```

```
        df_mean[col].fillna("NaN", inplace=True)
df_mean.isnull().sum()
```

```
C:\Users\Anshul Parkar\AppData\Local\Temp\
ipykernel_18016\3428114658.py:8: FutureWarning: A value is trying to
be set on a copy of a DataFrame or Series through chained assignment
using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never
work because the intermediate object on which we are setting values
always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try
using 'df.method({col: value}, inplace=True)' or df[col] =
df[col].method(value) instead, to perform the operation inplace on the
original object.


  df_mean[col].fillna("NaN", inplace=True)
C:\Users\Anshul Parkar\AppData\Local\Temp\
ipykernel_18016\3428114658.py:6: FutureWarning: A value is trying to
be set on a copy of a DataFrame or Series through chained assignment
using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never
work because the intermediate object on which we are setting values
always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try
using 'df.method({col: value}, inplace=True)' or df[col] =
df[col].method(value) instead, to perform the operation inplace on the
original object.


  df_mean[col].fillna(df_mean[col].mean(), inplace=True)
```

{"columns":[{"name":"index","rawType":"object","type":"string"},
{"name":"0","rawType":"int64","type":"integer"}],"ref":"90772c9e-f722-4155-9b3e-fbb7a025473d","rows":[["Loan_ID","0"],["Gender","0"],
["Married","0"],["Dependents","0"],["Education","0"],
["Self_Employed","0"],["ApplicantIncome","0"],
["CoapplicantIncome","0"],["LoanAmount","0"],["Loan_Amount_Term","0"],
["Credit_History","0"],["Property_Area","0"],
["Loan_Status","0"]],"shape":{"columns":1,"rows":13}}

```
df_mean
```

{"columns":[{"name":"index","rawType":"int64","type":"integer"},
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{"name":"Gender","rawType":"object","type":"string"},
{"name":"Married","rawType":"object","type":"string"},
{"name":"Dependents","rawType":"object","type":"string"},
{"name":"Education","rawType":"object","type":"string"},

{"name":"Self_Employed","rawType":"object","type":"string"},
{"name":"ApplicantIncome","rawType":"int64","type":"integer"},
{"name":"CoapplicantIncome","rawType":"float64","type":"float"},
{"name":"LoanAmount","rawType":"float64","type":"float"},
{"name":"Loan_Amount_Term","rawType":"float64","type":"float"},
{"name":"Credit_History","rawType":"float64","type":"float"},
{"name":"Property_Area","rawType":"object","type":"string"},
{"name":"Loan_Status","rawType":"object","type":"string"}],"ref":"33dd
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0","360.0","1.0","Rural","N"],
["2","LP001005","Male","Yes","0","Graduate","Yes","3000","0.0","66.0",
"360.0","1.0","Urban","Y"],["3","LP001006","Male","Yes","0","Not
Graduate","No","2583","2358.0","120.0","360.0","1.0","Urban","Y"],
["4","LP001008","Male","No","0","Graduate","No","6000","0.0","141.0","
360.0","1.0","Urban","Y"],
["5","LP001011","Male","Yes","2","Graduate","Yes","5417","4196.0","267
.0","360.0","1.0","Urban","Y"],["6","LP001013","Male","Yes","0","Not
Graduate","No","2333","1516.0","95.0","360.0","1.0","Urban","Y"],
["7","LP001014","Male","Yes","3+","Graduate","No","3036","2504.0","158
.0","360.0","0.0","Semiurban","N"],
["8","LP001018","Male","Yes","2","Graduate","No","4006","1526.0","168.
0","360.0","1.0","Urban","Y"],
["9","LP001020","Male","Yes","1","Graduate","No","12841","10968.0","34
9.0","360.0","1.0","Semiurban","N"],
["10","LP001024","Male","Yes","2","Graduate","No","3200","700.0","70.0
","360.0","1.0","Urban","Y"],
["11","LP001027","Male","Yes","2","Graduate","NaN","2500","1840.0","10
9.0","360.0","1.0","Urban","Y"],
["12","LP001028","Male","Yes","2","Graduate","No","3073","8106.0","200
.0","360.0","1.0","Urban","Y"],
["13","LP001029","Male","No","0","Graduate","No","1853","2840.0","114.
0","360.0","1.0","Rural","N"],
["14","LP001030","Male","Yes","2","Graduate","No","1299","1086.0","17.
0","120.0","1.0","Urban","Y"],
["15","LP001032","Male","No","0","Graduate","No","4950","0.0","125.0",
"360.0","1.0","Urban","Y"],["16","LP001034","Male","No","1","Not
Graduate","No","3596","0.0","100.0","240.0","0.8421985815602837","Urba
n","Y"],
["17","LP001036","Female","No","0","Graduate","No","3510","0.0","76.0"
,"360.0","0.0","Urban","N"],["18","LP001038","Male","Yes","0","Not
Graduate","No","4887","0.0","133.0","360.0","1.0","Rural","N"],
["19","LP001041","Male","Yes","0","Graduate","NaN","2600","3500.0","11
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Graduate","No","7660","0.0","104.0","360.0","0.0","Urban","N"],
["21","LP001046","Male","Yes","1","Graduate","No","5955","5625.0","315
.0","360.0","1.0","Urban","Y"],["22","LP001047","Male","Yes","0","Not

Graduate","No","2600","1911.0","116.0","360.0","0.0","Semiurban","N"],
["23","LP001050","NaN","Yes","2","Not
Graduate","No","3365","1917.0","112.0","360.0","0.0","Rural","N"],
["24","LP001052","Male","Yes","1","Graduate","NaN","3717","2925.0","15
1.0","360.0","0.8421985815602837","Semiurban","N"],
["25","LP001066","Male","Yes","0","Graduate","Yes","9560","0.0","191.0
","360.0","1.0","Semiurban","Y"],
["26","LP001068","Male","Yes","0","Graduate","No","2799","2253.0","122
.0","360.0","1.0","Semiurban","Y"],
["27","LP001073","Male","Yes","2","Not
Graduate","No","4226","1040.0","110.0","360.0","1.0","Urban","Y"],
["28","LP001086","Male","No","0","Not
Graduate","No","1442","0.0","35.0","360.0","1.0","Urban","N"],
["29","LP001087","Female","No","2","Graduate","NaN","3750","2083.0","1
20.0","360.0","1.0","Semiurban","Y"],
["30","LP001091","Male","Yes","1","Graduate","NaN","4166","3369.0","20
1.0","360.0","0.8421985815602837","Urban","N"],
["31","LP001095","Male","No","0","Graduate","No","3167","0.0","74.0","
360.0","1.0","Urban","N"],
["32","LP001097","Male","No","1","Graduate","Yes","4692","0.0","106.0"
,"360.0","1.0","Rural","N"],
["33","LP001098","Male","Yes","0","Graduate","No","3500","1667.0","114
.0","360.0","1.0","Semiurban","Y"],
["34","LP001100","Male","No","3+","Graduate","No","12500","3000.0","32
0.0","360.0","1.0","Rural","N"],
["35","LP001106","Male","Yes","0","Graduate","No","2275","2067.0","146
.41216216216216","360.0","1.0","Urban","Y"],
["36","LP001109","Male","Yes","0","Graduate","No","1828","1330.0","100
.0","342.0","0.0","Urban","N"],
["37","LP001112","Female","Yes","0","Graduate","No","3667","1459.0","1
44.0","360.0","1.0","Semiurban","Y"],
["38","LP001114","Male","No","0","Graduate","No","4166","7210.0","184.
0","360.0","1.0","Urban","Y"],["39","LP001116","Male","No","0","Not
Graduate","No","3748","1668.0","110.0","360.0","1.0","Semiurban","Y"],
["40","LP001119","Male","No","0","Graduate","No","3600","0.0","80.0","
360.0","1.0","Urban","N"],
["41","LP001120","Male","No","0","Graduate","No","1800","1213.0","47.0
","360.0","1.0","Urban","Y"],
["42","LP001123","Male","Yes","0","Graduate","No","2400","0.0","75.0",
"360.0","0.8421985815602837","Urban","Y"],
["43","LP001131","Male","Yes","0","Graduate","No","3941","2336.0","134
.0","360.0","1.0","Semiurban","Y"],
["44","LP001136","Male","Yes","0","Not
Graduate","Yes","4695","0.0","96.0","342.0","1.0","Urban","Y"],
["45","LP001137","Female","No","0","Graduate","No","3410","0.0","88.0"
,"342.0","1.0","Urban","Y"],
["46","LP001138","Male","Yes","1","Graduate","No","5649","0.0","44.0",
"360.0","1.0","Urban","Y"],
["47","LP001144","Male","Yes","0","Graduate","No","5821","0.0","144.0"

```
,"360.0","1.0","Urban","Y"],
["48","LP001146","Female","Yes","0","Graduate","No","2645","3440.0","1
20.0","360.0","0.0","Urban","N"],
["49","LP001151","Female","No","0","Graduate","No","4000","2275.0","14
4.0","360.0","1.0","Semiurban","Y"]],"shape":
{"columns":13,"rows":614}}
```



```python
na_variables = [ var for var in df.columns if df[var].isnull().mean()
> 0 ]
#for finding null values in cols
na_variables
```

```
['Gender',
 'Married',
 'Dependents',
 'Self_Employed',
 'LoanAmount',
 'Loan_Amount_Term',
 'Credit_History']
```

```python
# mean imputation
df1 = df
df1
missing_col = ["LoanAmount"]

for i in missing_col:
  df1.loc[df1.loc[:,i].isnull(),i]=df1.loc[:,i].mean()

df1
```

```
      Loan_ID  Gender Married Dependents     Education
Self_Employed  \
0    LP001002    Male      No          0      Graduate            No

1    LP001003    Male     Yes          1      Graduate            No

2    LP001005    Male     Yes          0      Graduate           Yes

3    LP001006    Male     Yes          0  Not Graduate            No

4    LP001008    Male      No          0      Graduate            No

..        ...     ...     ...        ...           ...           ...

609  LP002978  Female      No          0      Graduate            No

610  LP002979    Male     Yes         3+      Graduate            No

611  LP002983    Male     Yes          1      Graduate            No

612  LP002984    Male     Yes          2      Graduate            No
```

```
613  LP002990  Female      No          0      Graduate          Yes

     ApplicantIncome  CoapplicantIncome  LoanAmount  Loan_Amount_Term
\
0               5849                0.0  146.412162             360.0

1               4583             1508.0  128.000000             360.0

2               3000                0.0   66.000000             360.0

3               2583             2358.0  120.000000             360.0

4               6000                0.0  141.000000             360.0

..               ...                ...         ...               ...

609             2900                0.0   71.000000             360.0

610             4106                0.0   40.000000             180.0

611             8072              240.0  253.000000             360.0

612             7583                0.0  187.000000             360.0

613             4583                0.0  133.000000             360.0

     Credit_History Property_Area Loan_Status
0               1.0         Urban           Y
1               1.0         Rural           N
2               1.0         Urban           Y
3               1.0         Urban           Y
4               1.0         Urban           Y
..              ...           ...         ...
609             1.0         Rural           Y
610             1.0         Rural           Y
611             1.0         Urban           Y
612             1.0         Urban           Y
613             0.0     Semiurban           N

[614 rows x 13 columns]
```

```
# median imputation
df2=df
for i in missing_col:
  df2.loc[df2.loc[:,i].isnull(),i]=df2.loc[:,i].median()

df2
```

```
      Loan_ID  Gender Married Dependents     Education
Self_Employed  \
0    LP001002    Male      No          0      Graduate            No

1    LP001003    Male     Yes          1      Graduate            No

2    LP001005    Male     Yes          0      Graduate           Yes

3    LP001006    Male     Yes          0  Not Graduate            No

4    LP001008    Male      No          0      Graduate            No

..        ...     ...     ...        ...           ...           ...

609  LP002978  Female      No          0      Graduate            No

610  LP002979    Male     Yes         3+      Graduate            No

611  LP002983    Male     Yes          1      Graduate            No

612  LP002984    Male     Yes          2      Graduate            No

613  LP002990  Female      No          0      Graduate           Yes


     ApplicantIncome  CoapplicantIncome  LoanAmount  Loan_Amount_Term
\
0               5849                0.0  146.412162             360.0

1               4583             1508.0  128.000000             360.0

2               3000                0.0   66.000000             360.0

3               2583             2358.0  120.000000             360.0

4               6000                0.0  141.000000             360.0

..               ...                ...         ...               ...

609             2900                0.0   71.000000             360.0

610             4106                0.0   40.000000             180.0

611             8072              240.0  253.000000             360.0

612             7583                0.0  187.000000             360.0

613             4583                0.0  133.000000             360.0


     Credit_History Property_Area Loan_Status
0               1.0         Urban           Y
```

```
1                 1.0         Rural           N
2                 1.0         Urban           Y
3                 1.0         Urban           Y
4                 1.0         Urban           Y
..                ...           ...         ...
609               1.0         Rural           Y
610               1.0         Rural           Y
611               1.0         Urban           Y
612               1.0         Urban           Y
613               0.0     Semiurban           N

[614 rows x 13 columns]

# Mode imputation

df4 = df
df4
missing_col = ["LoanAmount"]

for i in missing_col:
  df4.loc[df4.loc[:,i].isnull(),i]=df4.loc[:,i].mode()

df4

      Loan_ID  Gender Married Dependents     Education
Self_Employed  \
0    LP001002    Male      No          0      Graduate            No

1    LP001003    Male     Yes          1      Graduate            No

2    LP001005    Male     Yes          0      Graduate           Yes

3    LP001006    Male     Yes          0  Not Graduate            No

4    LP001008    Male      No          0      Graduate            No

..        ...     ...     ...        ...           ...           ...

609  LP002978  Female      No          0      Graduate            No

610  LP002979    Male     Yes         3+      Graduate            No

611  LP002983    Male     Yes          1      Graduate            No

612  LP002984    Male     Yes          2      Graduate            No

613  LP002990  Female      No          0      Graduate           Yes


     ApplicantIncome  CoapplicantIncome  LoanAmount  Loan_Amount_Term
\
```

```
0                5849              0.0  146.412162             360.0
1                4583           1508.0  128.000000             360.0
2                3000              0.0   66.000000             360.0
3                2583           2358.0  120.000000             360.0
4                6000              0.0  141.000000             360.0
..                ...              ...         ...               ...
609              2900              0.0   71.000000             360.0
610              4106              0.0   40.000000             180.0
611              8072            240.0  253.000000             360.0
612              7583              0.0  187.000000             360.0
613              4583              0.0  133.000000             360.0

     Credit_History Property_Area Loan_Status
0               1.0         Urban           Y
1               1.0         Rural           N
2               1.0         Urban           Y
3               1.0         Urban           Y
4               1.0         Urban           Y
..              ...           ...         ...
609             1.0         Rural           Y
610             1.0         Rural           Y
611             1.0         Urban           Y
612             1.0         Urban           Y
613             0.0     Semiurban           N

[614 rows x 13 columns]
```

```python
#categorical to numerical

from sklearn.preprocessing import OrdinalEncoder

data=df
oe =OrdinalEncoder()
result = oe.fit_transform(data)
print(result)
```

```
[[  0.   1.   0. ...   1.   2.   1.]
 [  1.   1.   1. ...   1.   0.   0.]
 [  2.   1.   1. ...   1.   2.   1.]
```

```
 ...
 [611.   1.   1. ...   1.   2.   1.]
 [612.   1.   1. ...   1.   2.   1.]
 [613.   0.   0. ...   0.   1.   0.]]
```

```
#random sample
df5=df
df5['LoanAmount'].dropna().sample(df5['LoanAmount'].isnull().sum(),ran
dom_state=0)
df5
```

```
      Loan_ID  Gender Married Dependents     Education 
Self_Employed  \
0    LP001002    Male      No          0      Graduate            No

1    LP001003    Male     Yes          1      Graduate            No

2    LP001005    Male     Yes          0      Graduate           Yes

3    LP001006    Male     Yes          0  Not Graduate            No

4    LP001008    Male      No          0      Graduate            No

..        ...     ...     ...        ...           ...           ...

609  LP002978  Female      No          0      Graduate            No

610  LP002979    Male     Yes         3+      Graduate            No

611  LP002983    Male     Yes          1      Graduate            No

612  LP002984    Male     Yes          2      Graduate            No

613  LP002990  Female      No          0      Graduate           Yes


     ApplicantIncome  CoapplicantIncome  LoanAmount  Loan_Amount_Term 
\
0               5849                0.0  146.412162             360.0

1               4583             1508.0  128.000000             360.0

2               3000                0.0   66.000000             360.0

3               2583             2358.0  120.000000             360.0

4               6000                0.0  141.000000             360.0

..               ...                ...         ...               ...

609             2900                0.0   71.000000             360.0
```

```
610              4106                0.0   40.000000            180.0

611              8072              240.0  253.000000            360.0

612              7583                0.0  187.000000            360.0

613              4583                0.0  133.000000            360.0


     Credit_History Property_Area Loan_Status
0               1.0         Urban           Y
1               1.0         Rural           N
2               1.0         Urban           Y
3               1.0         Urban           Y
4               1.0         Urban           Y
..              ...           ...         ...
609             1.0         Rural           Y
610             1.0         Rural           Y
611             1.0         Urban           Y
612             1.0         Urban           Y
613             0.0     Semiurban           N

[614 rows x 13 columns]
```

```
# frequent category imputation
df6=df
m= df6["Gender"].mode()
m=m.tolist()

frq_imp = df6["Gender"].fillna(m[0])
frq_imp.unique()
```

```
array(['Male', 'Female'], dtype=object)
```

```
#regression imputation
from sklearn.linear_model import LinearRegression
lr = LinearRegression()
df1=df[["CoapplicantIncome","LoanAmount"]]


# col=df1["LoanAmount"].dropna()
# df1.head()
testdf = df1[df1['LoanAmount'].isnull()==True]
testdf
traindf = df1[df1['LoanAmount'].isnull()==False]
traindf


lr.fit(traindf['LoanAmount'],traindf['CoapplicantIncome'])
# testdf.drop("LoanAmount",axis=1,inplace=True)
```

```python
# testdf
pred = lr.predict(testdf)
# testdf['LoanAmount']= pred
```

```
     CoapplicantIncome  LoanAmount
0                  0.0  146.412162
1               1508.0  128.000000
2                  0.0   66.000000
3               2358.0  120.000000
4                  0.0  141.000000
..                 ...         ...
609                0.0   71.000000
610                0.0   40.000000
611              240.0  253.000000
612                0.0  187.000000
613                0.0  133.000000

[614 rows x 2 columns]
```